Derivation of the Effective Connectivity $J_{\text{eff}}(x, a, b)$

This note derives the Jacobian of the x-dynamics with respect to x, treating the adaptation variables a and the synaptic depression variables b as constants (i.e., "frozen" at a given time t_1). The result is an effective connectivity matrix $J_{\text{eff}}(x, a, b)$ that depends on the current state. Rather that considering the entire Jacobian in which the adaptation variables are included in the state, this approach helps the reader visualize how adaptation can be viewed as modulating connectivity, which is a concept investigated in the manuscript.

1. Model definition

We consider a network of N neurons (indices $i, j = 1, \ldots, N$) with the following equations:

1.1 x-dynamics

The membrane / rate state variable x_i evolves as

$$\frac{dx_i}{dt} = \frac{-x_i + u_i + \sum_{j=1}^{N} w_{ij}\, b_j\, r_j}{\tau_d},$$

where

- τ_d is the (global) decay time constant,
- u_i is an external input (ignored when computing the Jacobian),
- w_{ij} is the (structural) weight from neuron j to neuron i,
- b_j is the synaptic depression variable for neuron j,
- r_j is the firing rate of neuron j.

1.2 Nonlinearity and adaptation

The firing rate r_i is given by

$$r_i = \phi\left(x_i - a_{0_i} - c\sum_{k=1}^{K} a_{ik}\right),$$

where

- $\phi(\cdot)$ is a static nonlinearity (e.g., sigmoid),
- a_{0_i} is an offset parameter (e.g., preferred input or threshold shift),
- c is an adaptation gain,
- a_{ik} are adaptation state variables for neuron i and component $k = 1, \ldots, K$.

The synaptic depression variable $b_i \in [0, 1]$ multiplies r_i in the dynamics equation, so that the effective synaptic output of neuron j is $b_j r_j$.

The dynamics of a_{ik} and b_i are not needed to compute J_{eff}, since we explicitly treat a and b as constants during this derivation.

2. Vector-matrix notation

Let

- $x = (x_1, \ldots, x_N)^T$,
- $r = (r_1, \ldots, r_N)^T$,
- $u = (u_1, \ldots, u_N)^T$,
- $b = (b_1, \ldots, b_N)^T$,
- $W = (w_{ij}) \in \mathbb{R}^{N \times N}$,
- $B = \text{diag}(b_1, \ldots, b_N)$,

- $\mathbf{1}$ be the N-vector of ones.

The recurrent input term can be written as

$$\left(\sum_{j=1}^{N} w_{ij}\, b_j\, r_j \right)_i = (WBr)_i.$$

Thus, the x-dynamics in vector form is

$$\dot{x} = \frac{-x + u + WBr}{\tau_d}.$$

3. Jacobian of r with respect to x

We treat a_{ik} and b_i as constants. For each neuron i,

$$r_i = \phi\left(x_i - a_{0_i} - c \sum_{k=1}^{K} a_{ik} \right).$$

Define the constant

$$\text{const}_i = a_{0_i} + c \sum_{k=1}^{K} a_{ik}.$$

Then

$$r_i = \phi\left(x_i - \text{const}_i \right).$$

The partial derivative of r_i with respect to x_j is

$$\frac{\partial r_i}{\partial x_j} = \phi'\left(x_i - \text{const}_i \right)\, \delta_{ij},$$

where δ_{ij} is the Kronecker delta.

Define the gain vector

$$g_i = \phi'\left(x_i - a_{0_i} - c \sum_{k=1}^{K} a_{ik} \right),$$

and the corresponding diagonal matrix

$$G = \text{diag}(g_1, \ldots, g_N).$$

In matrix form, the Jacobian of r with respect to x is

$$\frac{\partial r}{\partial x} = G.$$

4. Jacobian of $\dot{x}$ with respect to x

We now compute

$$J_{\text{eff}}(x, a, b) = \frac{\partial \dot{x}}{\partial x}.$$

Recall

$$\dot{x} = \frac{-x + u + WBr}{\tau_d}.$$

4.1 Contribution from the leak term

For the leak term $-x/\tau_d$,

$$\frac{\partial}{\partial x_j}\left(-\frac{x_i}{\tau_d}\right) = -\frac{1}{\tau_d}\,\delta_{ij}.$$

Thus, in matrix form this contributes

$$-\frac{1}{\tau_d}I,$$

where I is the $N \times N$ identity matrix.

4.2 Contribution from the recurrent term

For the recurrent term $(1/\tau_d)\,WBr$,

$$\frac{\partial}{\partial x_j}\left(\frac{1}{\tau_d}\sum_{k=1}^{N} w_{ik}\, b_k\, r_k\right) = \frac{1}{\tau_d}\sum_{k=1}^{N} w_{ik}\, b_k\, \frac{\partial r_k}{\partial x_j}.$$

Using $\partial r_k/\partial x_j = g_k \delta_{kj}$, we get

$$\frac{\partial}{\partial x_j}\left(\frac{1}{\tau_d}\sum_{k=1}^{N} w_{ik}\, b_k\, r_k\right) = \frac{1}{\tau_d}\, w_{ij}\, b_j\, g_j.$$

So the entries of the Jacobian are

$$J_{\text{eff},ij} = \frac{\partial \dot{x}_i}{\partial x_j} = -\frac{1}{\tau_d}\delta_{ij} + \frac{1}{\tau_d}\, w_{ij}\, b_j\, g_j.$$

In matrix form, note that $(WBG)_{ij} = \sum_k w_{ik}\, b_k\, g_k\, \delta_{kj} = w_{ij}\, b_j\, g_j$, since B and G are both diagonal. Therefore

$$J_{\text{eff}}(x, a, b) = \frac{1}{\tau_d}\left(-I + WBG\right).$$

Since B and G are both diagonal, their product commutes and can be combined into a single diagonal matrix $\tilde{G} = BG$:

$$\tilde{G} = \operatorname{diag}\Big(b_i\, \phi'\big(x_i - a_{0_i} - c\sum_{k=1}^{K} a_{ik}\big)\Big).$$

5. Summary

- The effective connectivity Jacobian for the x-dynamics, treating a and b as constants and ignoring external input u in the derivative, is

$$J_{\text{eff}}(x,a,b) = \frac{1}{\tau_d}\left(-I + W\tilde{G}\right),$$

where

$$\tilde{G} = \operatorname{diag}\Big(b_i\,\phi'\big(x_i - a_{0_i} - c\sum_{k=1}^{K} a_{ik}\big)\Big).$$

- Entrywise, this is

$$J_{\text{eff},ij} = -\frac{1}{\tau_d}\delta_{ij} + \frac{1}{\tau_d} w_{ij}\, b_j\, \phi'\left(x_j - a_{0_j} - c\sum_{k=1}^{K} a_{jk}\right).$$